JAIN DEEMED-TO-BE UNIVERSITY
FACULTY OF ENGINEERING AND TECHNOLOGY

ADVANCED COMPUTER NETWORKS

CourseCode:23CSE503

L:T:P:2-0-2-2

Prerequisite(s): Data Communication and Networks

Credits:03

Contact Hours: 60

Course Objectives

The objective of the course is to

1. To Build an understanding of the fundamental concepts of computer networking
2. To Familiarize the student with the basic taxonomy and terminology of the computer networking area
3. To Understand state-of-the-art in network protocols, architectures, and applications
4. To Identify a router as a computer with an operating system(OS) and hardware designed for the routing process

Course Outcomes

At the end of the course, students will be able to

CourseOutcomes	**Description**	**Bloom'sTaxonomyLevel**
CO1	Explain the application layer of the OSI model and TCP/IP.	Understanding(2)
CO2	Discuss the different Transport layer protocols	Understanding(2)
CO3	Apply the networking layer concepts for IPV4 and sub netting.	Applying(3)
CO4	Implement the different dynamic routing protocols like RIP, EIGRP and OSPF.	Applying(3)
CO5	Discuss the basic protocols, VLANs, VTP, WAN, ATM in networks.	Understanding(2)

CO6	Discuss the contemporary issues in wifi and 802.11networks.	Understanding(2)

CourseContents

Module 1 **[12Hours]**

Networking Concepts: Layered operation, Protocol Suites. **Application Layer:** Application layer functions and Application Layer protocols, HTTP, FTP, SMTP, Electronic mail, Domain name system DNS, World Wide Web.

Module2 **[12Hours]**

The transport layer: The Transport service, End-to-End Protocols, Elements of Transport protocols, Simple, transport protocol Transmission Control Protocol(TCP), Internet transport protocols, TCP Connection Establishment, TCP Flow Control, TCP Congestion Control,Weighted Fair Queuing(WFQ), Random Early detection(RED), Internet transport protocols TCP, User Datagram Protocol(UDP)

Module3 **[12Hours]**

Network Layer: Static and Dynamic IP addressing, Public and Private Networks, IPV4, Limitations of current IPV4, Internet Protocol Version 6 (IPv6) features, DHCP, Routing and Packet Forwarding, Static Routing, **Introduction to Dynamic Routing Protocols**, Distance Vector Routing Protocol, RIPv1,Variable length sub-net mask-VLSM, Class less inter domain routing-CIDR, Routing Table- A Closer Look, EIGRP, Link State Routing Protocol, OSPF, Overview of BGP

Module4 **[12Hours]**

Link Layer and LAN: Multiple access Networks, ALOHSs, CSMA, CSMA/CD and CSMA/CA, Stability in Slotted ALOHA, Bayesian Algorithm,- Splitting Algorithms Tree and First Come First Serve FCFS Scheduling Switch,VLANs,VLANTrunkProtocol-VTP,Implement Spanning Tree Protocol, Inter-VLAN Routing, Wireless Router Services in Converged WAN, Point to Point Protocol,Frame Relay.
ATM Networks: Introduction, ATM, The ATMAAL Layer Protocols, Historical perspective, protocol architecture.

Module5 **[12Hours]**

Wi-Fi: Topologies of wireless networks, world of Radio frequency, Framework and operation of 802.11networks, Physical design of wireless networks, Introduction to antennas, security of 802.11 networks, Bluetooth.

Multimedia:-RTP, Enterprise Network Security Assess Control List

Textbook

1. "ComputerNetworking,ATop-DownApproachFeaturingtheInternet",JamesF.KuroseandKeithW.Ross,6th Ed,Addison-Wesley,2012.
2. w

   AdvancedComputerNetorkPaperback–Import,11March2020https://www.amazon.in/Advanced-Computer-Network-Rahul-Sharma/dp/6202511842

Reference Books

1. "Data and Computer Communications", William Stallings,10th Edition, Pearson Prentice Hall,2013.
2. "Data Communications and Networking", Behrouz A. Forouzan, 4th Edition, McGraw-Hill, 2006
3. "Computer Networks–A System Approach", L.L.Peterson and B.S.Davie, 5th Edition, Elsevier, 2011
4. "Computer Networks", Andrew S.Tanenbaum, 5th Edition, Prentice Hall of India, 2010